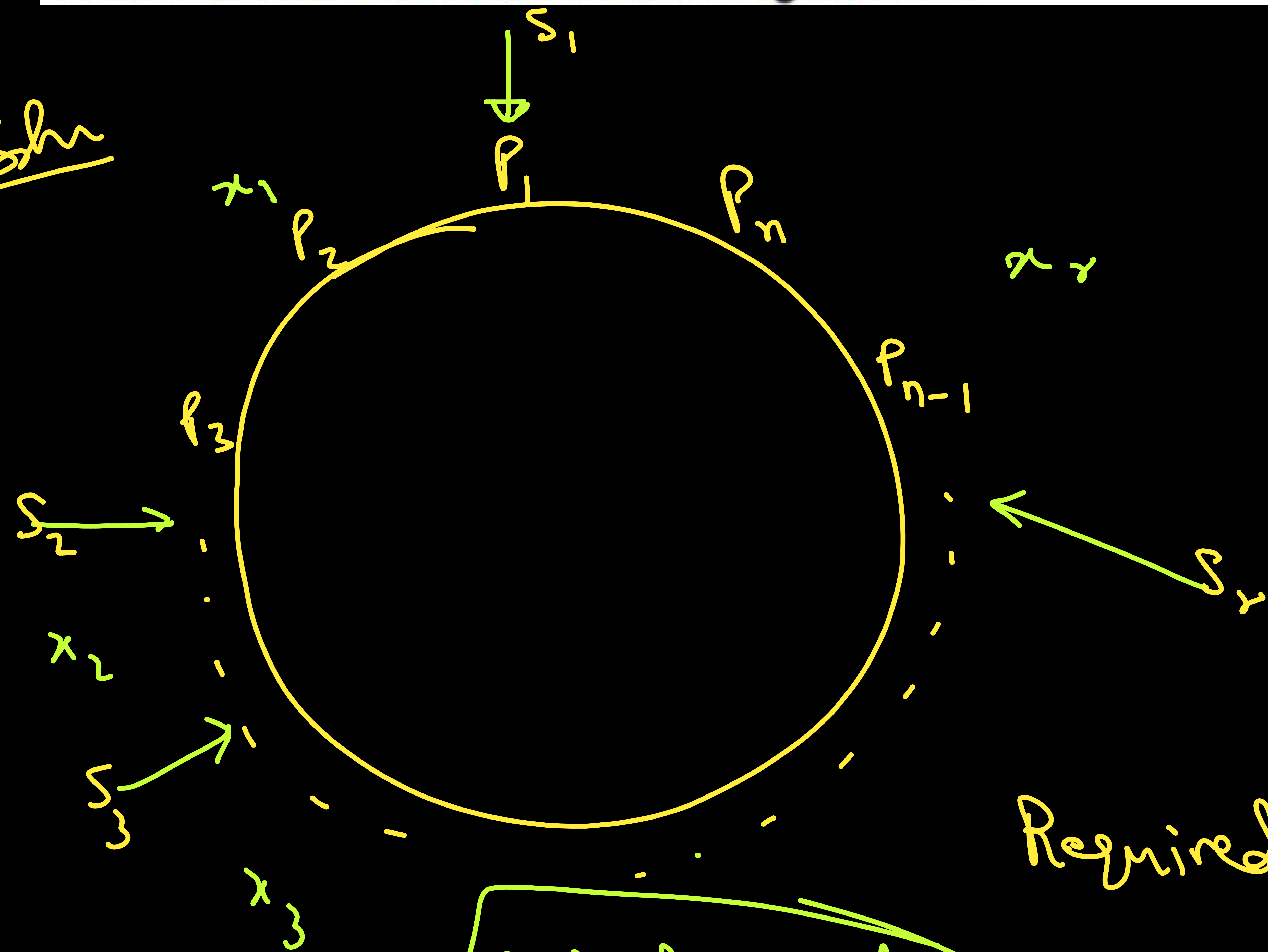


Ques

In how many ways we can select r persons out of n persons sitting around a round table such that no two of them are neighbours.

Soln



$P_1, P_2, P_3, \dots, P_{n-3}$
 $\swarrow$
 r persons

$$x_1 + x_2 + x_3 + \dots + x_r = n - r$$

$$x_1 \geq 1, x_2 \geq 1, \dots, x_r \geq 1$$



$$x'_1 + x'_2 + \dots + x'_r = n - 2r$$

$$x'_i \geq 0$$

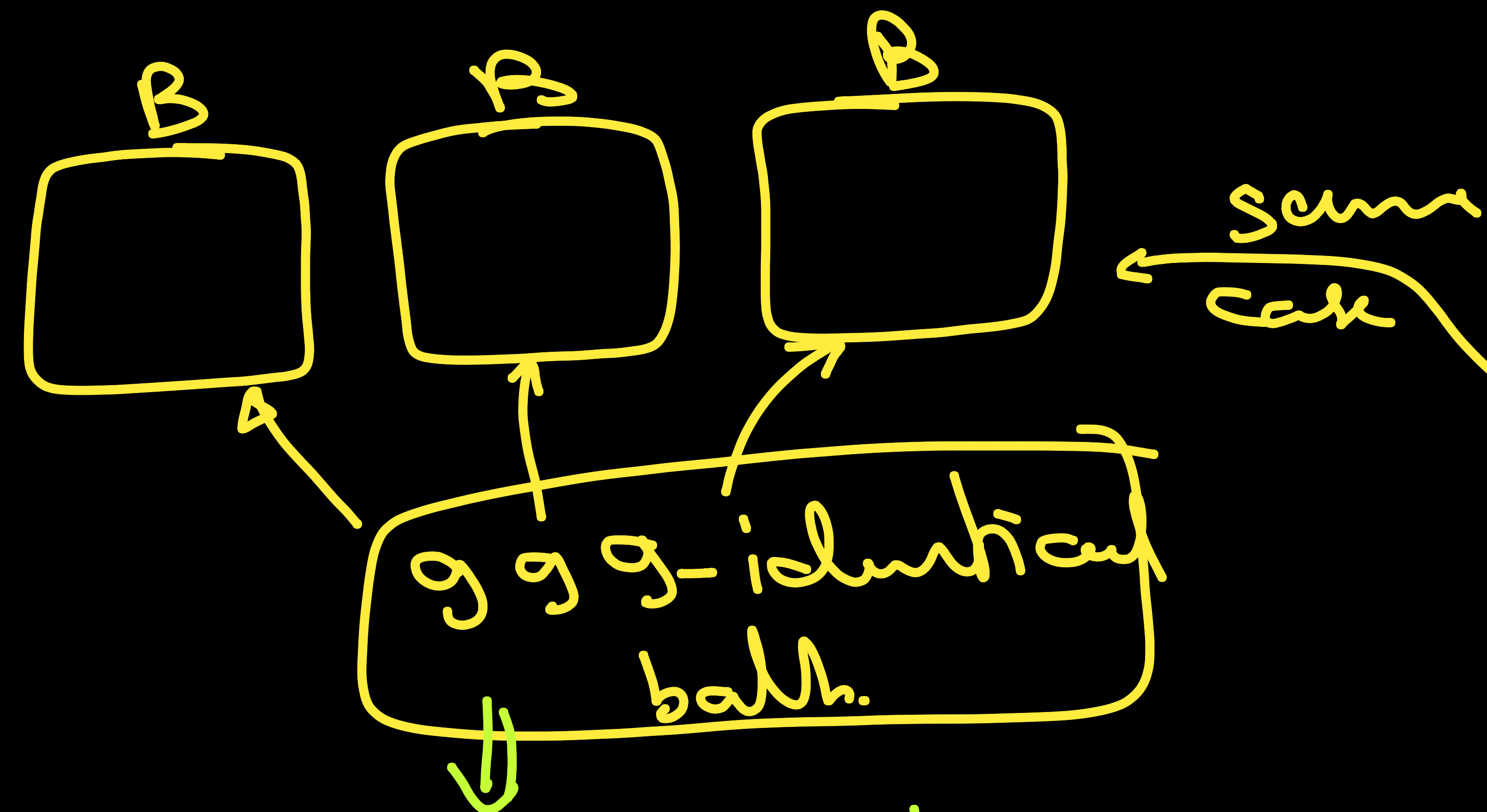
$$\text{N.O.S.} = \binom{n-2r+r-1}{r-1} = \binom{n-r-1}{r-1}$$

$$\text{Required Ans} = \binom{n-r-1}{r-1}$$

Ques

In how many ways we can distribute 999 identical balls in 3 identical boxes.

Sol



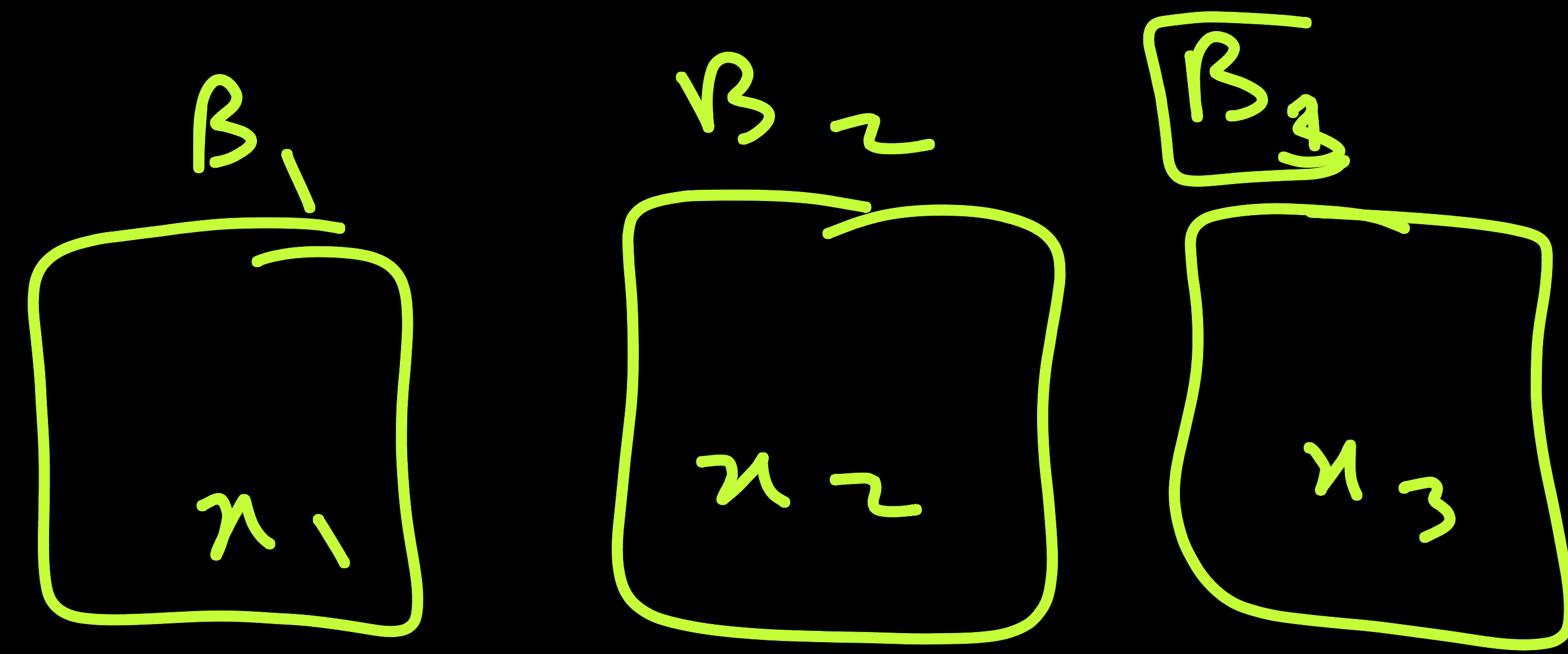
How many in a box.
What is the combination? X

$$x_1 + x_2 + x_3 = 999$$

$$\begin{aligned} \hookrightarrow \text{N.O.S.} &= \binom{999+2}{2} = \binom{1001}{2} \\ &= \frac{1001 \times 1000}{2} = 1001 \times 500 \\ &= \underline{\underline{500500}} \end{aligned}$$

B_1	B_2	B_3
x_1	x_2	x_3
$\begin{array}{r} 99 \\ 450 \\ \hline 549 \end{array}$	$\begin{array}{r} 450 \\ 99 \\ \hline 549 \end{array}$	$\begin{array}{r} 450 \\ 450 \\ 99 \\ \hline 999 \end{array}$
333	333	333
100	800	99
100	99	800
99	100	800
99	800	100

6 cases



$$x_1 + x_2 + x_3 = 999 \longrightarrow \text{NOS.} = 500500$$



Required Ans.

$$\begin{aligned}
 &= 1 + \frac{n_1}{3} + \frac{n_2}{3!} \\
 &= 1 + 499 + \frac{499002}{6} \\
 &= 500 + 83167 \\
 &= \boxed{83667} \quad \underline{A}
 \end{aligned}$$

For n_1

$$\begin{aligned}
 &x_1 = x_2 \neq x_3 \\
 &\rightarrow x_1 + x_1 + x_3 = 999 \\
 &\quad 2x_1 + x_3 = 999 \\
 &\rightarrow \text{N.O.S.} = 500 - 1 \\
 &\quad = 499
 \end{aligned}$$

$$n_1 = 30499 = 1497$$

$$\begin{aligned}
 &x_1 = x_2 \neq x_3 \\
 &x_1 = x_3 \neq x_2 \\
 &x_2 = x_3 \neq x_1
 \end{aligned}$$

$$\begin{aligned}
 n_2 &= 500500 - 1 - 1497 \\
 &= \frac{500500}{1498} \\
 &\quad \underline{499002}
 \end{aligned}$$

Ques

Find the total number of seven digit numbers $x_1x_2x_3x_4x_5x_6x_7$ having the property that $x_1 \leq x_2 < x_3 < x_4 \leq x_5 < x_6 < x_7$.

Soln

$$\begin{aligned} & x_1 \leq x_2 < x_3 < x_4 \leq x_5 < x_6 < x_7 \\ & \rightarrow x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7 \rightarrow {}^9C_7 \\ & \rightarrow x_1 = x_2 < x_3 < x_4 < x_5 < x_6 < x_7 \rightarrow {}^9C_6 \\ & \rightarrow x_1 < x_2 < x_3 < x_4 = x_5 < x_6 < x_7 \rightarrow {}^9C_6 \\ & \rightarrow x_1 = x_2 < x_3 < x_4 = x_5 < x_6 < x_7 \rightarrow {}^9C_5 \\ & \hline & \left[{}^9C_7 + {}^9C_6 + {}^9C_6 + {}^9C_5 \right] \\ & \quad \left[{}^{10}C_7 + {}^{10}C_6 \right] = \boxed{{}^{11}C_7} \quad \text{Ans} \end{aligned}$$

One

Find the possible number of ordered pairs (m, n, p) such that $1 \leq m \leq 100$, $1 \leq n \leq 50$, $1 \leq p \leq 25$ and $2^m + 2^n + 2^p$ is divisible by 3.

Ans

$$\begin{aligned} 2^m + 2^n + 2^p &= (3-1)^m + (3-1)^n + (3-1)^p \\ &= (3k_1 + (-1)^m) + (3k_2 + (-1)^n) + (3k_3 + (-1)^p) \\ &= 3(k_1 + k_2 + k_3) + (-1)^m + (-1)^n + (-1)^p \end{aligned}$$

According to question either all 3- of m, n, p must be even

⊙ all 3- must be odd.

Hence, Required number of ordered pairs = $50 \cdot 25 \cdot 12 + 50 \cdot 25 \cdot 13$
 $= 50 \times 25 \times 25$

How many 5 letters word can be formed using the letters of word 'MANAGEMENT' such that if any two alike letters are there then they are always together.

M M
A A
N N
E
T

5-letter.

All diff

2-A + 3-Diff

2-Alike + 2-Alike
+ 1-Diff

$${}^6C_5 \cdot 5$$

$${}^4C_1 \cdot {}^5C_3 \cdot 4$$

$${}^4C_2 \cdot {}^4C_1 \cdot 3$$

$$6 \cdot 5 + 4 \times 10 \cdot 4 + 6 \times 4 \cdot 3$$

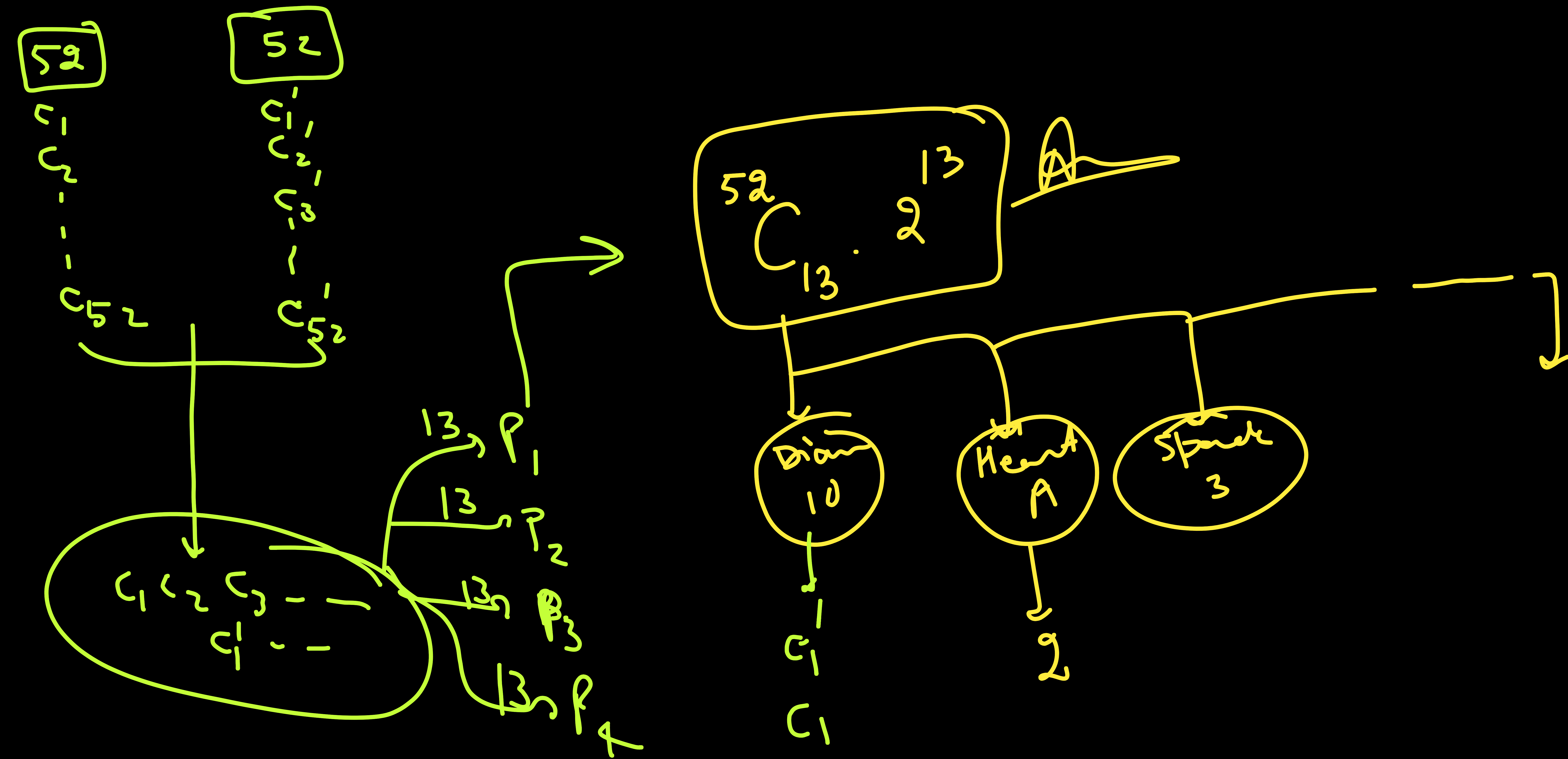
$$= 14 \cdot 5 + 6 \cdot 4 = 76 \cdot 4 = \underline{\underline{76 \times 24}}$$

✓

Que

Two different packs of card are shuffled together. Cards are distributed equally among 4 players, each getting 13 card. In how many ways can a player get his cards if no two cards are from the same suit and with the same denomination?

Sol



There are mn balls, m each of n different colours (including Red, Blue, and Black). Find the number of ways in which they can be arranged in a row so that no black ball appears before any red ball and no red ball appears before any Blue ball. Discuss the cases when balls of same colour are identical and also when balls of same colour are distinct.

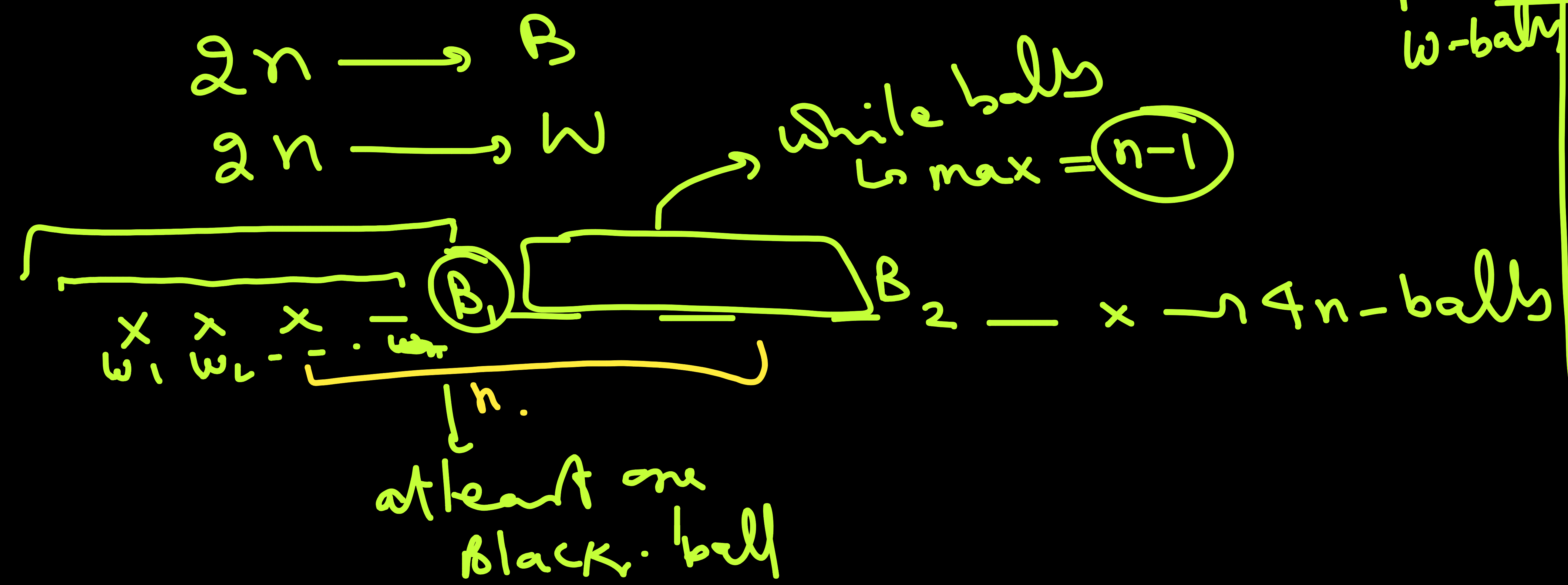
$$G_n \rightarrow B_1, B_2, \dots, B_m$$
$$R.N.O.C. = {}^{mn}C_{3m} \cdot \frac{(mn-3m)!}{(m!)^{n-3}}$$

Balls of same colour are diff.

C-2 $R.N.O.C. = {}^{mn}C_{3m} \cdot (\underline{1m})(\underline{1m})(\underline{1m}) \cdot \underline{(mn-3m)}$

Ques

In how many ways $2n$ identical black and $2n$ identical white balls can be arranged in a row such that on picking any n consecutive balls from the arrangement we get at least a black ball.



$$R.A_n = {}^{3n}C_n - (n+1) \cdot {}^{2n}C_n + \frac{n(n+1)}{2}$$

No. of W-balls $\rightarrow x_1, x_2, x_3, \dots, x_{n+1}$

R.N.O.W. = No. of integral solns of $x_1 + x_2 + x_3 + \dots + x_{n+1} = 2n$.

$0 \leq x_i \leq n-1$.

$$= \text{coeff of } x^{2n} \text{ in } \left((1+x+\dots+x^{n-1})^{n+1} \right)$$

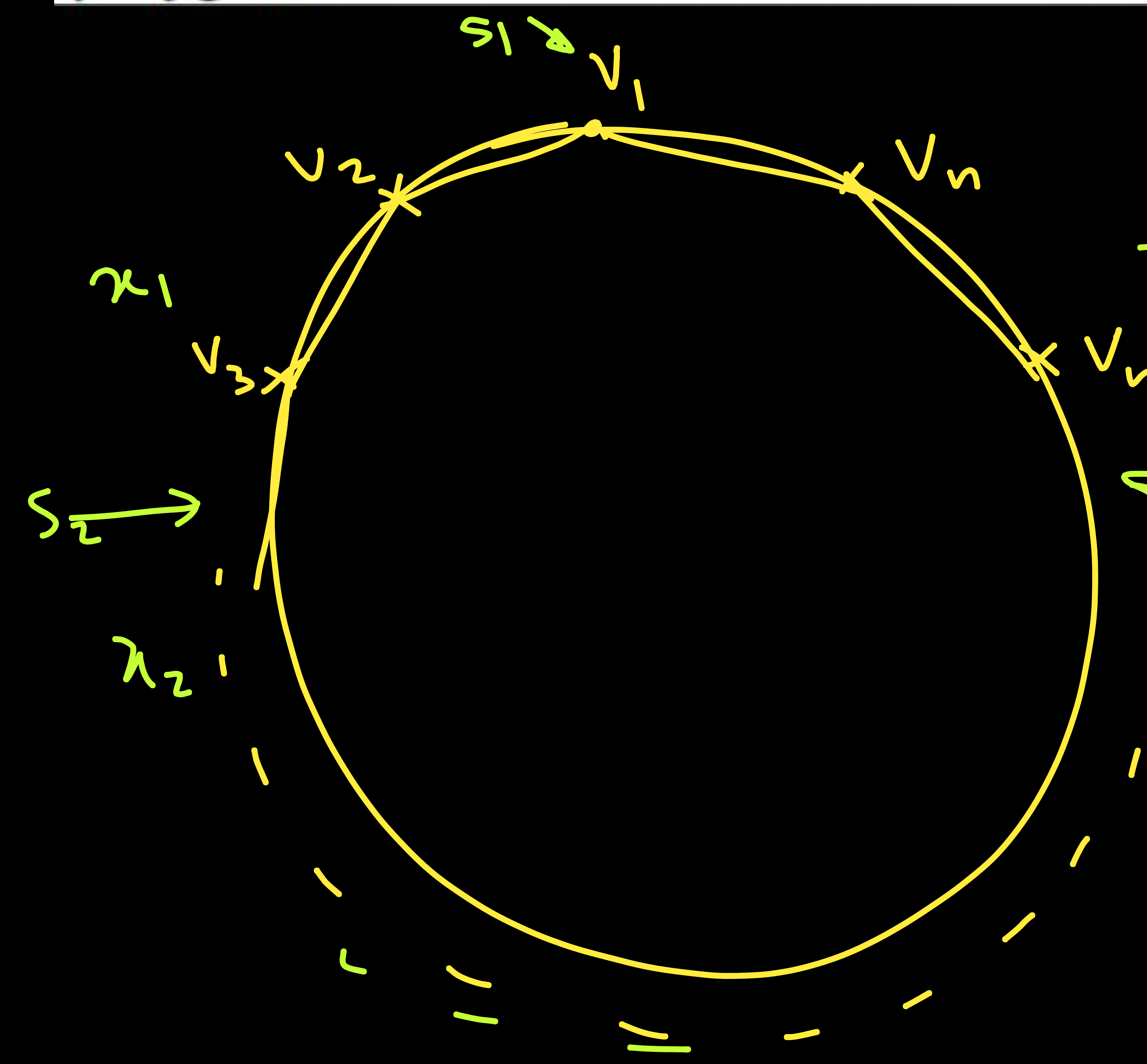
$$= \text{coeff of } x^{2n} \text{ in } \left((1-x^n)^{n+1} (1-x)^{-(n+1)} \right)$$

$$= \dots \left((1 - (n+1)x^n + \frac{(n+1)n}{2}x^{2n} - \dots) \cdot (1-x)^{-(n+1)} \right)$$

$$= {}^{2n+n}C_n - (n+1) {}^{n+n}C_n + \frac{(n+1)n}{2} \cdot 1$$

Ques

r -sided polygons are formed by joining the vertices of an n -sided polygon. Find the number of polygons that can be formed, none of whose sides coincide with those of the n -sided polygon.



$(n-1)$

$$x_1 + x_2 + \dots + x_r = n - r$$

$$x_i \geq 1$$

$\Downarrow$

$$x'_1 + x'_2 + \dots + x'_r = n - 2r$$

$$x'_i \geq 0$$

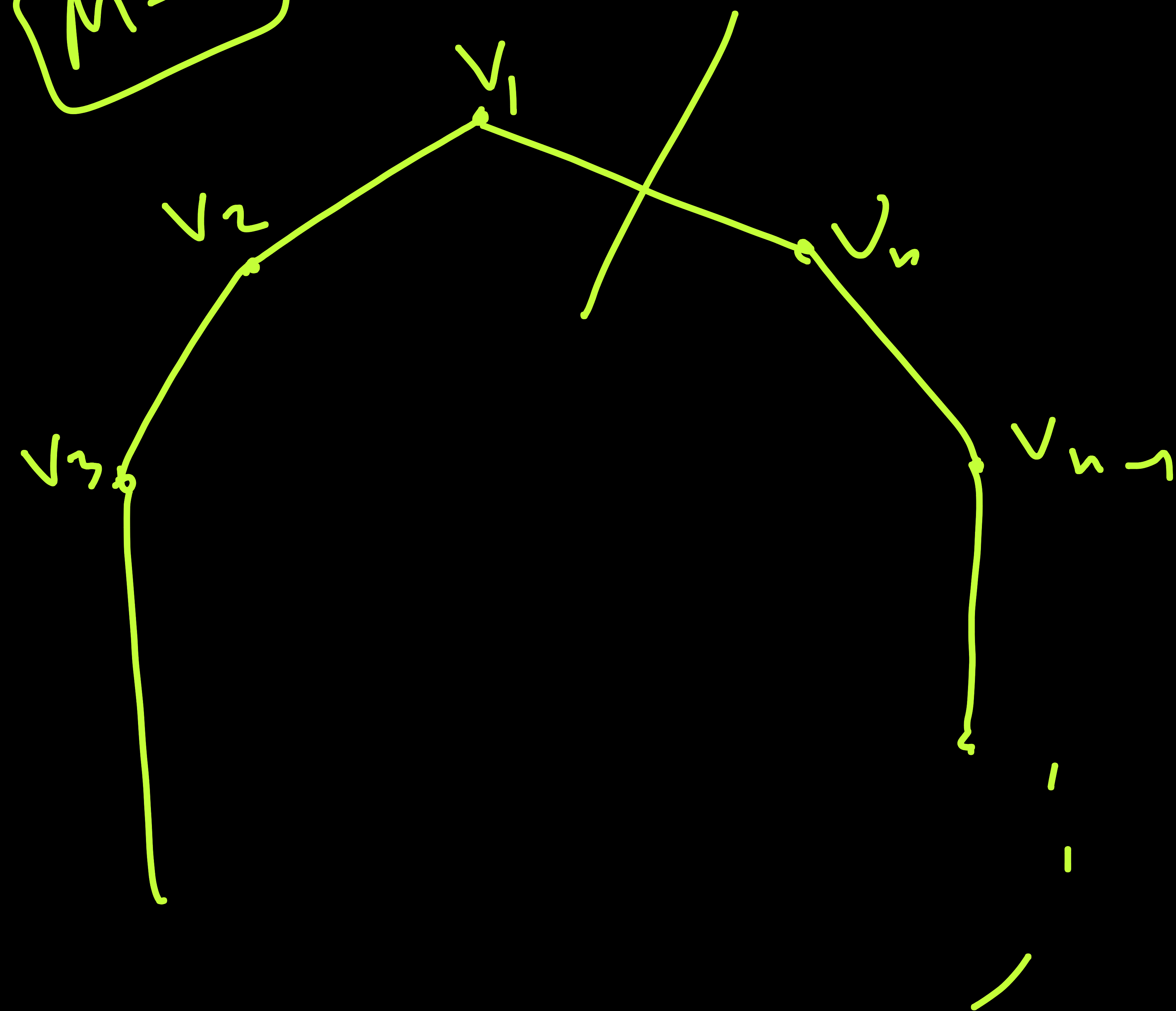
N.O.S. =

$$C_{n-2r+r-1}^{n-2r+r-1} = C_{n-r-1}^{n-r-1}$$

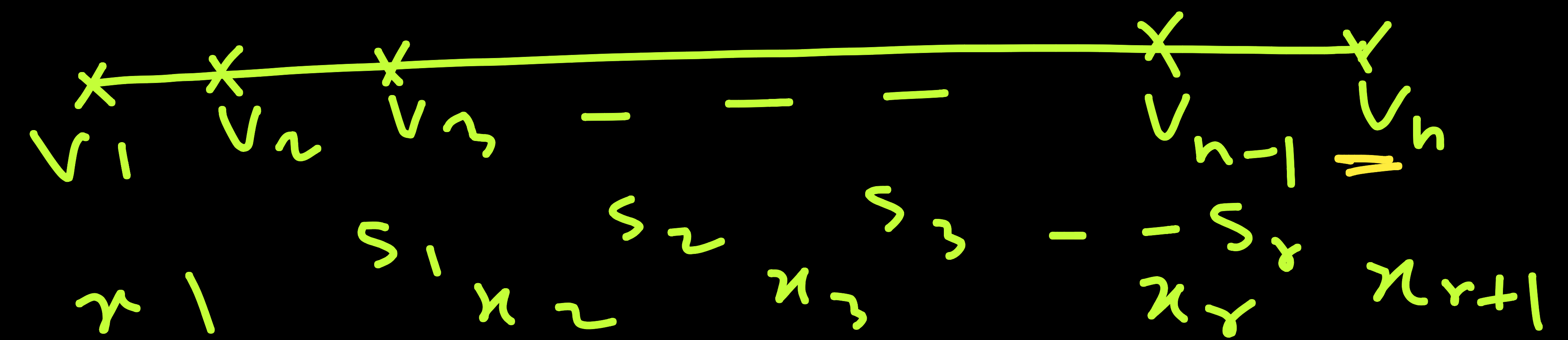
$$\text{Required no. of polygons} = \frac{n \cdot C_{n-r-1}^{n-r-1}}{r}$$

Ans

$M-2$



$\underline{\underline{C-1}}$ None of $v_1 \text{ \textcircled{&}} v_n$ is selected.



$$x_1 + x_2 + \dots + x_{r+1} = n - r$$

$$x_1, x_2, x_3, \dots, x_{r+1} \geq 1$$

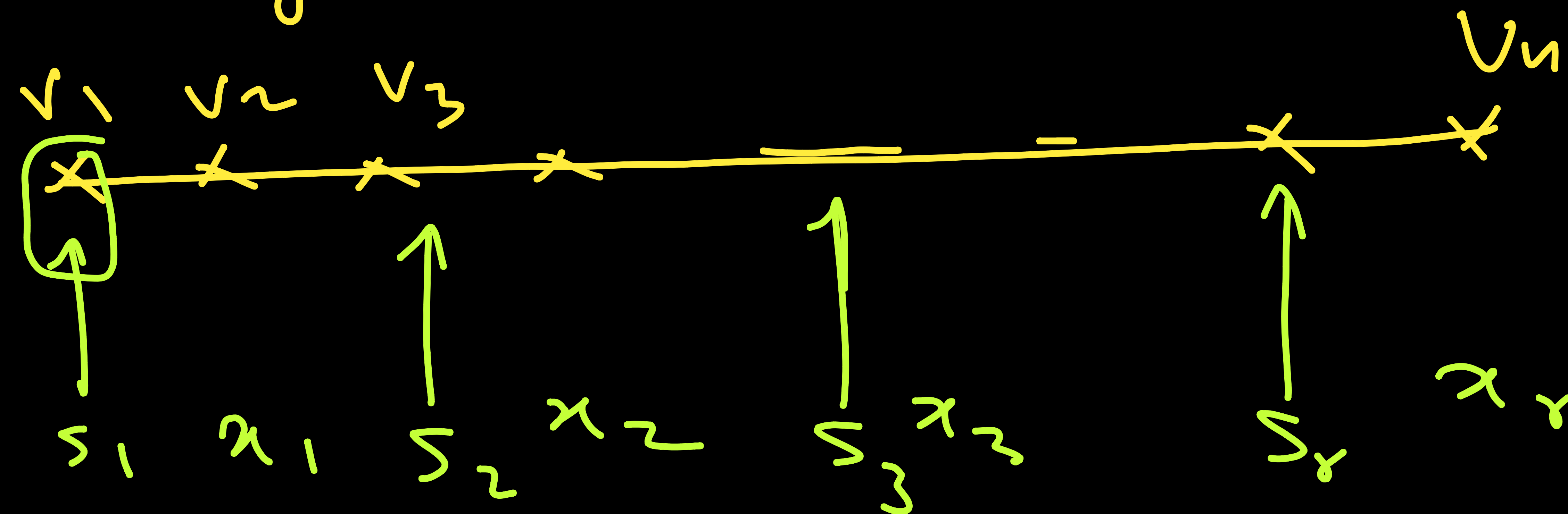
$$\Downarrow$$

$$x'_1 + x'_2 + \dots + x'_{r+1} = n - 2r - 1$$

$$x'_i \geq 0.$$

$$\text{R.N.O.W.} = \binom{n-2r-1+r}{r} = \binom{n-r-1}{r}.$$

C-2 Exactly one out of $V_1 \oplus V_n$ is selected.



$$x_1 + x_2 + \dots + x_r = n - r$$

$$x_i \geq 1$$

$$\Rightarrow \text{N.O. S.} = {}^{n-r+r-1}C_{r-1} = {}^{n-r-1}C_{r-1}$$

$$\text{R.N.O.W.} = 2 \cdot {}^{n-r-1}C_{r-1}$$

$$\text{Final } A_y = {}^{n-r-1}C_r + 2 \cdot {}^{n-r-1}C_{r-1}$$